



Deconvolution of transcriptomic changes caused by drug perturbation using matrix factorization

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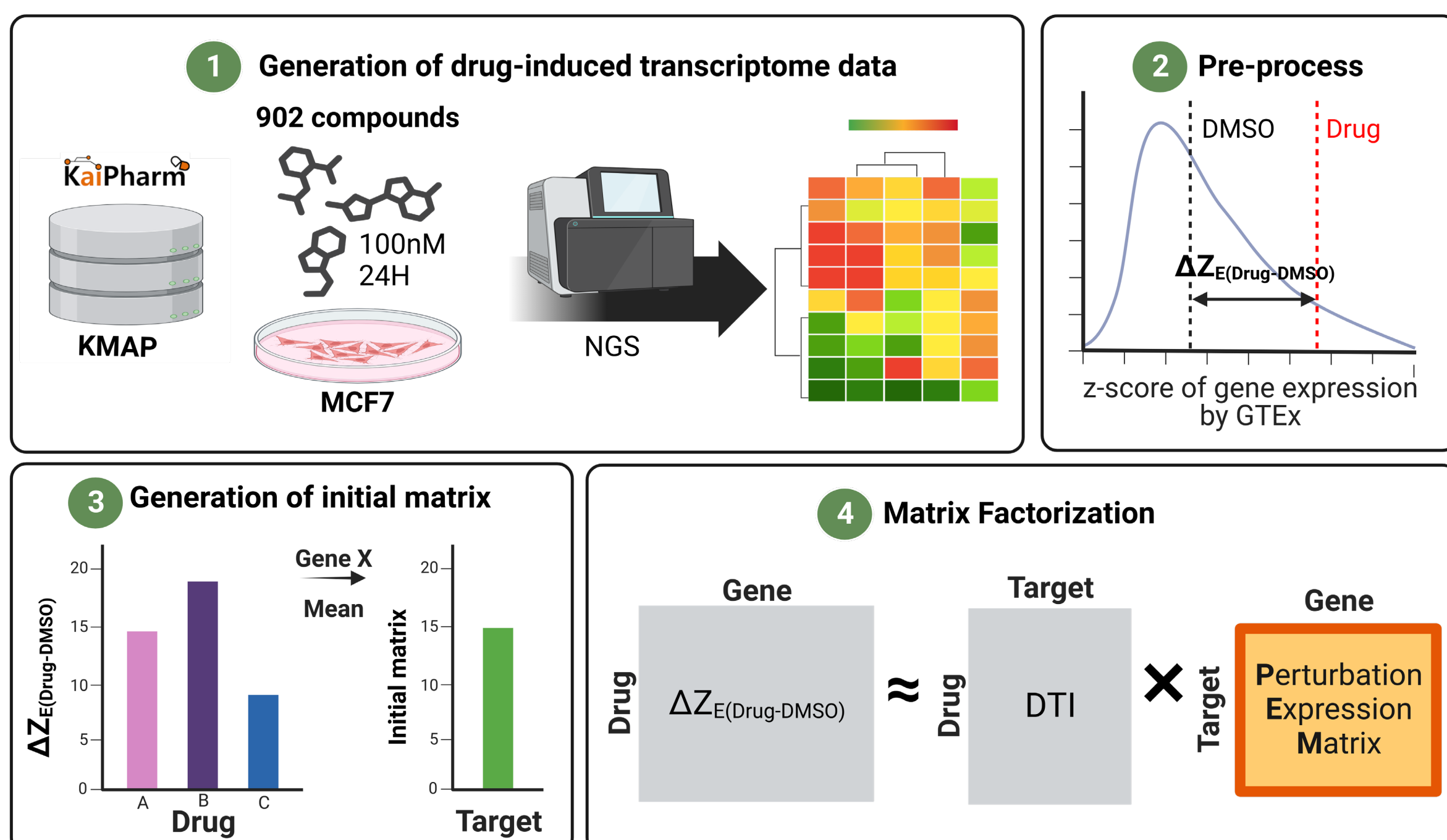
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Abstract

In the process of drug discovery and development, understanding drug mode-of-action (MoA) is an essential but challenging task. Analyzing gene expression changes resulting from drug perturbations offers a systematic and comprehensive means to understand a drug's impact on cellular processes. However, due to the mixed effects by multiple targets (i.e. on-targets and off-targets), it is difficult to clearly decompose the transcriptomic effects by each target of a drug. Here, we propose a matrix factorization model to uncover the relationship between the changes of gene expression and their causal targets. We deconvolute transcriptomic changes caused by drug multi-target perturbation down to a single target. Our model takes a vector of target profile as input and generates normalized transcriptomic changes as output, resulting in a perturbed expression matrix (PEM) of hidden relationship between targets and the resulting expression changes. As a result, the model predicted known drug target interaction with median AUROC=0.85. Moreover, we confirmed that the predicted perturbation profiles of drug targets exhibit high similarity to the differentially expressed genes induced by the corresponding drugs. Notably, the consideration of multi-target effects yielded the highest degree of similarity in comparison to the assessment of individual targets. By clustering targets based on their PEM values, each cluster was enriched in distinct pathways, indicating that the PEM method effectively separates patterns related to the mode of action (MoA) of drug targets. Furthermore, we applied our model to predict potential therapeutic targets for triple-negative breast cancer, successfully identifying four targets that are already under investigation in clinical studies, in addition to uncovering a novel target.

Materials & Methods



1. Materials

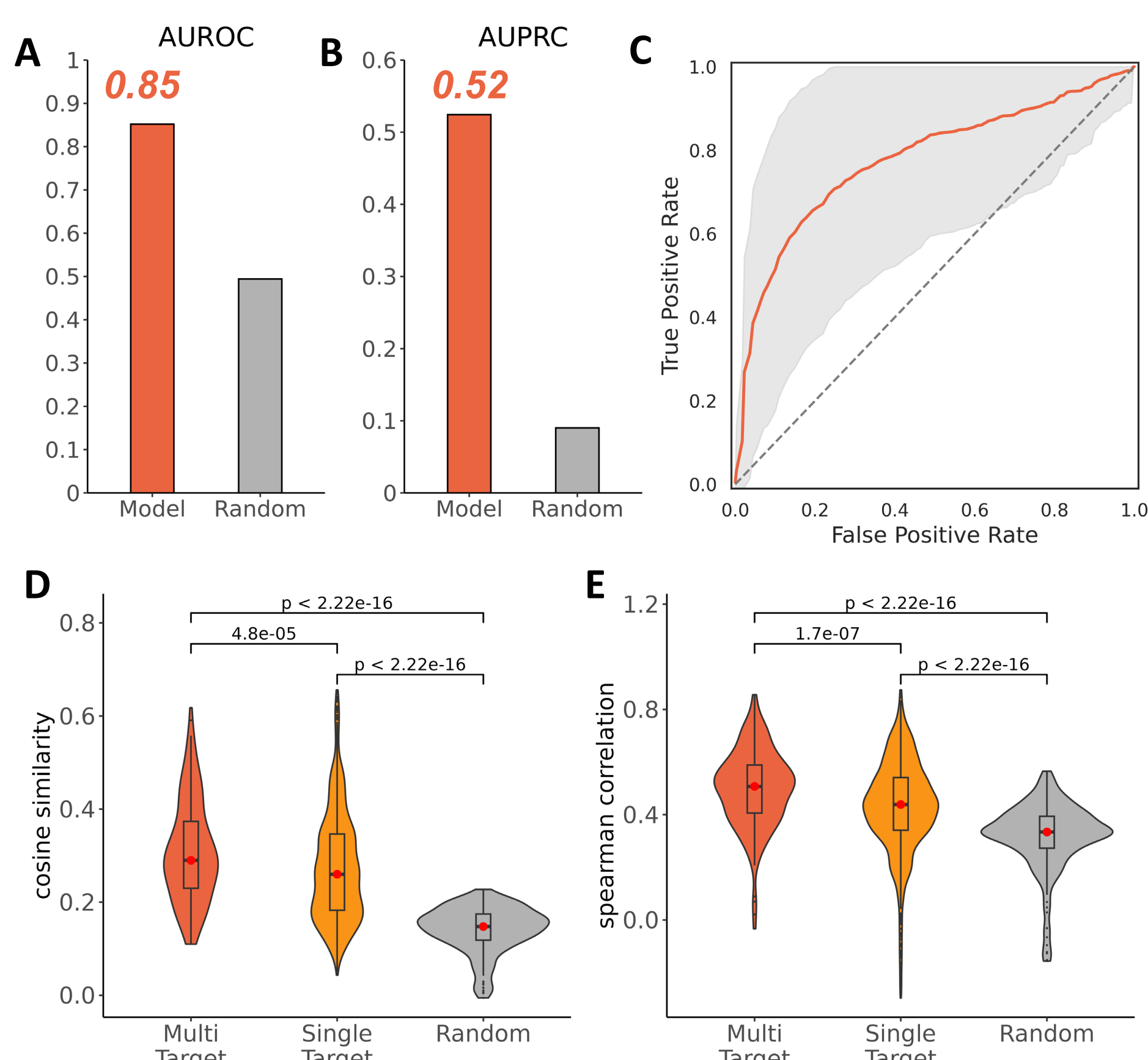
We collected drug-target interaction information from several public databases: ChEMBL (Gaulton et al., 2012), DrugBank (Wishart et al., 2018), IUPHAR, DrugCentral (Ursu et al., 2017), Therapeutic Targets Database (Zhou et al., 2022), and STITCH (Szklarczyk et al., 2016). After mapping compounds into Pubchem ID and proteins with UniprotKB ID, we selected targets annotated by more than 3 drugs and compounds that have less than 110 known target proteins. For the drug perturbed transcriptome dataset, we used KMAP, in-house data of Kaipharm. The dataset encompasses NGS data of MCF7 cells perturbed by 902 different compounds. Dataset resulted in the identification of 836 distinct targets and 9134 DTIs.

2. Methods

We normalized the transcript-per-million (TPM) counts of KMAP data with the GTEx database (Lonsdale et al., 2013). We calculated the difference in z-scores between drug-perturbed gene expression and expression treated with DMSO as the vehicle control and used this matrix as delta-normalized drug-induced gene expression ($\Delta Z_{E(Drug-DMSO)}$). We adopted matrix factorization to derive a PEM that incorporates expected change of gene expression caused by individual drug targets. Model takes a binary vector of target profile as input and generates normalized transcriptomic changes by drug perturbation. By comparing the cosine similarity of the degree of change by the drug and the PEM of each target in the weight, we get the rank of drug targets. The greater the cosine similarity, the greater the likelihood of the target. We evaluated the performance of the model using the metrics of AUROC and AUPRC.

Results

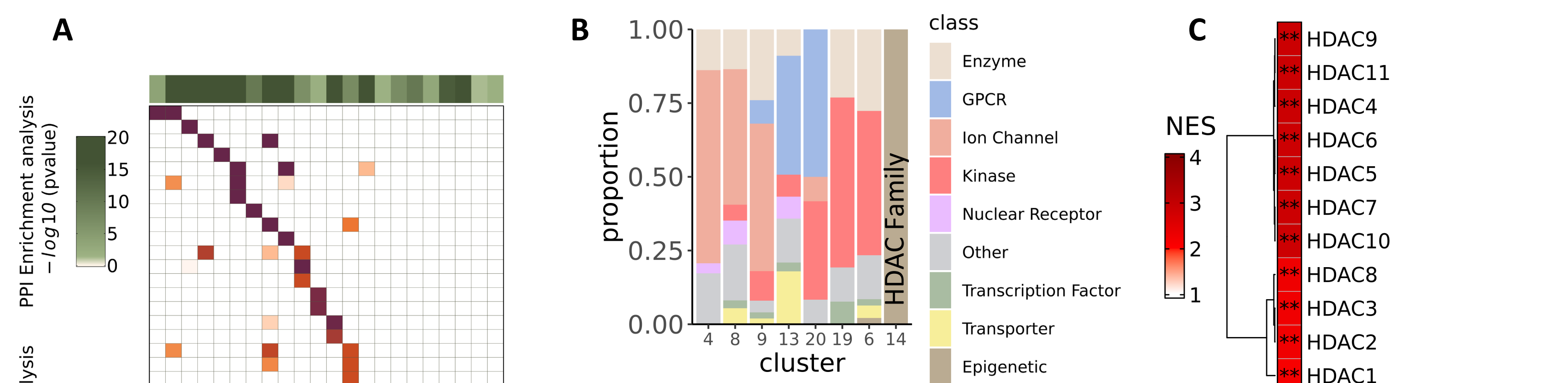
I. PEM predicts known targets of the test-set drugs with a median AUROC 0.85.



A, B, C. Our model achieved a median AUROC as 0.852 and a median AUPRC as 0.524. To investigate the presence of bias within the weight matrix, we introduced randomness into the weight matrix by shuffling all of its elements and subsequently evaluated the model's performance under the same methodology and showed equivalent to random chance.

D, E. The cumulative weight matrix of all known targets demonstrated the highest similarity to gene expression data, with a median cosine similarity of 0.29 and a median spearman correlation of 0.51. These outcomes signify that the drug targets works synergistically and collectively and it is elaborated as gene expression when perturbed by drug.

IV. The clusters identified by PEM display distinct enrichment in pathways.



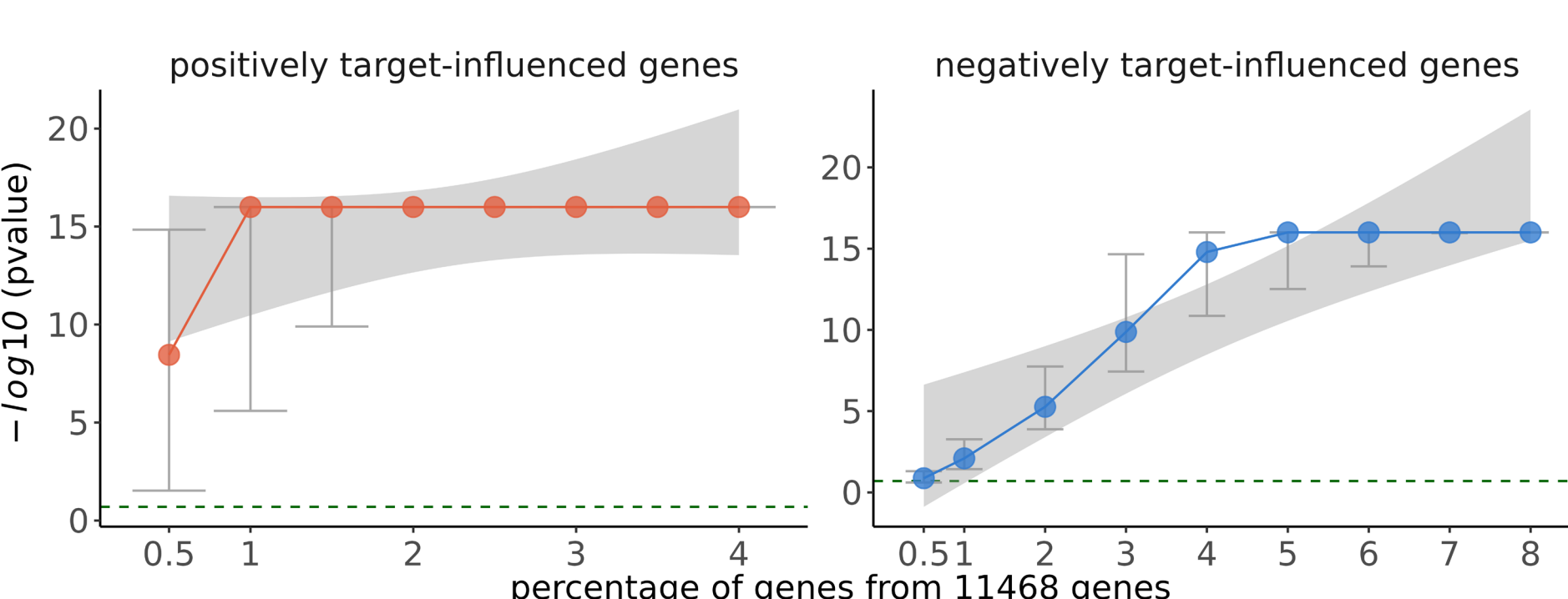
We selected only the top 10% genes with high median absolute deviation and performed consensus clustering with the k-means algorithm. It result 22 clusters.

A. All 22 clusters exhibited significant PPI interactions ($P < 0.05$) among the targets within clusters. Additionally, the targets within the same cluster exhibited distinct enrichment in specific pathways

B. Within each cluster, we observed a proportion of protein classes annotated by Pharos. Particularly, 11 histone deacetylase (HDAC) proteins is aggregated in cluster 14.

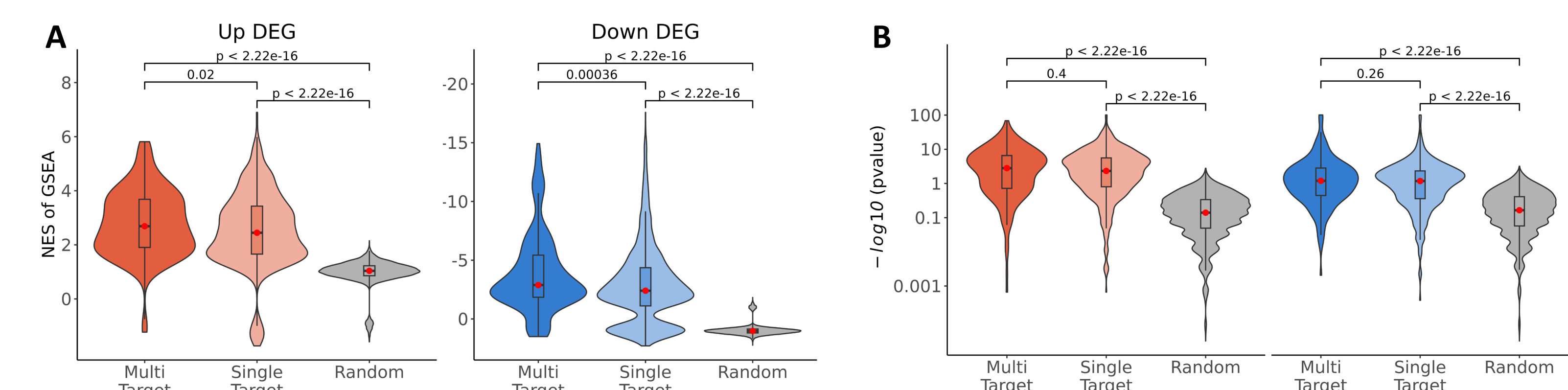
C. PEM of HDAC family proteins are all significantly enriched in Reactome 'HDACS mediated Histone Deacetylation' pathway annotated genes.

II. PEM shows a significant interaction between the target and its most likely influenced genes.



Targets and their 0.5% PTIGs and NTIGs demonstrated slightly significant interaction. Importantly, as the number of genes increased, it showed a increasing significant level of interaction. The shaded gray region in the figure represents the expected model calculated using linear regression.

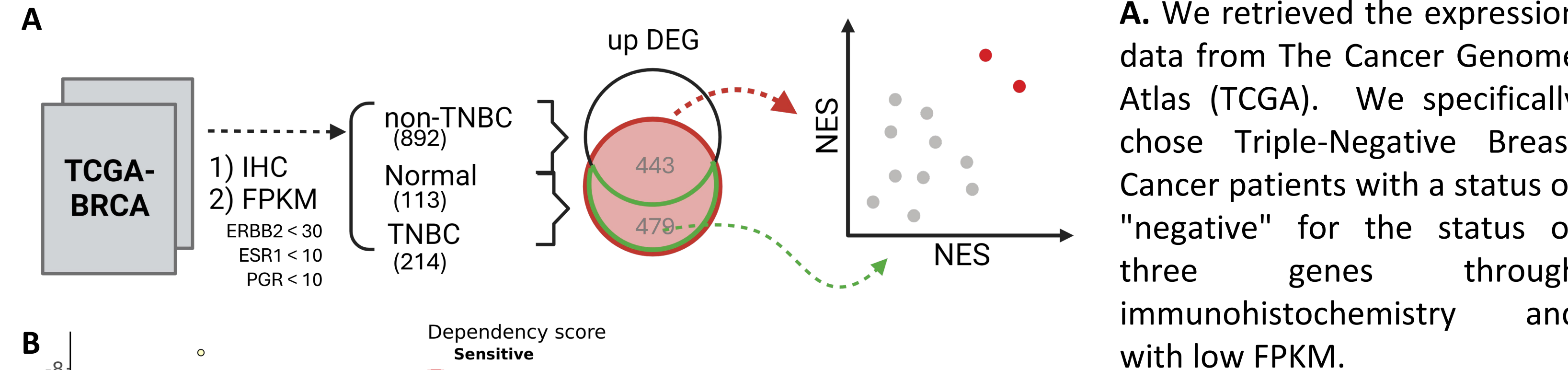
III. PEM exhibits a statistically significant enrichment with the DEGs induced by the drug.



We conducted a comparative analysis to examine the overlap between the genes predicted as most likely influenced by targets based on the PEM and differentially expressed genes (DEG) selected by DESeq2 induced by the drug. Similarly, considering multi target effect showed the most significant overlap with DEGs.

A. Gene Set Enrichment Analysis (GSEA) B. Hypergeometric test

V. PEM predicts targets for TNBC currently under clinical study, as well as novel targets.



A. We retrieved the expression data from The Cancer Genome Atlas (TCGA). We specifically chose Triple-Negative Breast Cancer patients with a status of "negative" for the status of three genes through immunohistochemistry and with low FPKM.

We aimed to identify targets exhibiting a significant negative Normalized Enrichment Score (NES) with genes that were overexpressed in the tumor, indicating their potential to downregulate these overexpressed genes. To specify DEGs unique to TNBC, as distinct from other types of breast cancer, we employed two subsets of DEGs (1) DEG compared to TNBC over normal (2) DEGs found in TNBC versus normal, excluding those in non-TNBC versus normal.

B. We identified 21 drug targets that exhibited significant NES with both subsets of DEGs. Among these targets, four (USP1, RRM1, CDA, NT5E) have inhibitors that are already in clinical trials for TNBC. Notably, ISM3091, a novel selective USP1 initiated its first-in-human phase 1 study on BRCA-deficient BC in July 2023. This highlights the potential of these four novel targets for therapeutic intervention. Experimental validation of these novel target inhibitors is proceeding as planned.

Summary

- We constructed a Perturbation Expression Matrix (PEM) using matrix factorization, integrating a drug-target interaction binary matrix with normalized gene expression data induced by drug perturbation.
- Clusters identified by PEM may share common patterns of MoA, reinforcing the utility of the PEM method in dissecting the functional relationships among drug targets.
- Through the prediction of therapeutic targets for TNBC with the potential to restore the disease state to normal by downregulating overexpressed genes, PEM successfully identified targets that are currently under investigation in clinical studies, underlining the clinical relevance and applicability of this approach in the field of cancer therapeutics.